SOAR PLAYBOOK DEVELOPMENT REPORT

Avinash Shankar P T
Cyber Security Intern

1. Objective

The objective of this task was to design and test a Security Orchestration, Automation, and Response (SOAR) playbook to automate incident response for phishing alerts. The playbook focuses on detecting malicious IPs, validating them using threat intelligence, blocking them, and creating an incident record.

2. Tools Used

- Wazuh – Alert generation and detection
- AlienVault OTX – IP reputation check
- TheHive – Case management (simulated)
- Linux (iptables) – IP blocking mechanism

3. Playbook Workflow

The following automated workflow was designed:
Wazuh Alert → Extract IP → Check OTX → Block IP → Create Case

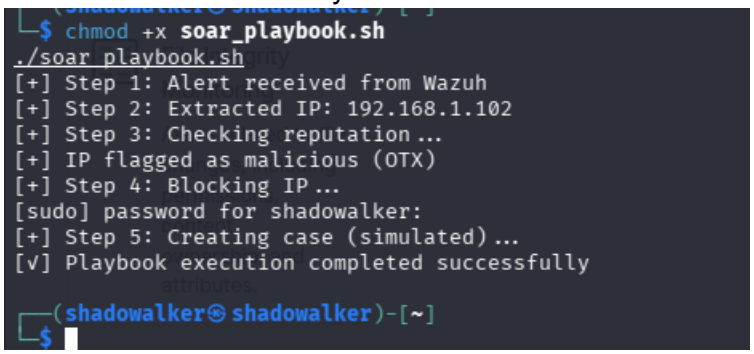
Steps:

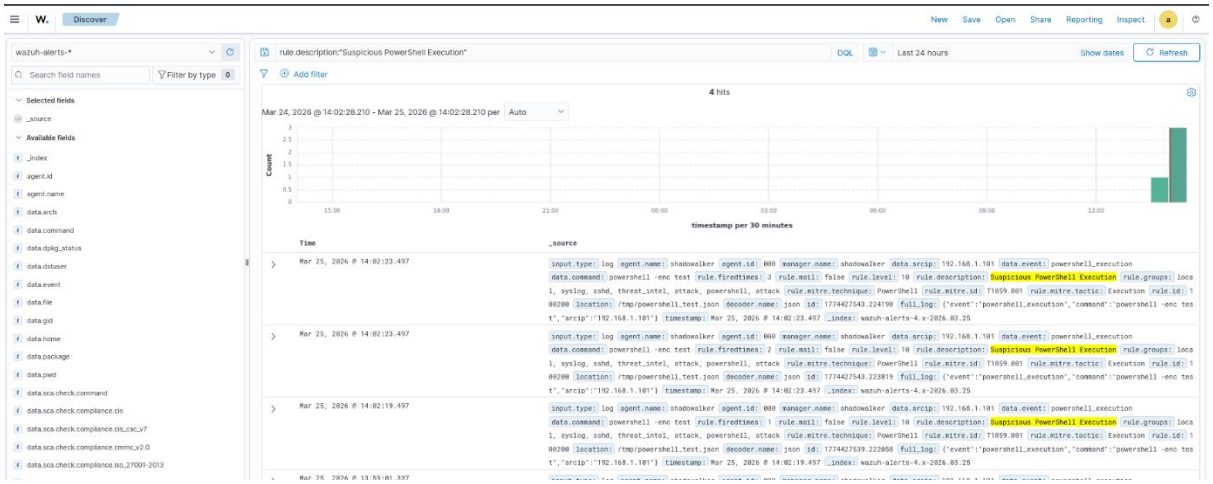
1. Detect phishing alert from Wazuh
2. Extract source IP address
3. Check IP reputation using OTX
4. If malicious → block IP using firewall
5. Create incident case for tracking

4. Playbook Implementation

4.1 Alert Simulation

A phishing alert was generated using:
logger "Phishing email detected from 192.168.1.102"
The alert was successfully detected in the Wazuh dashboard.



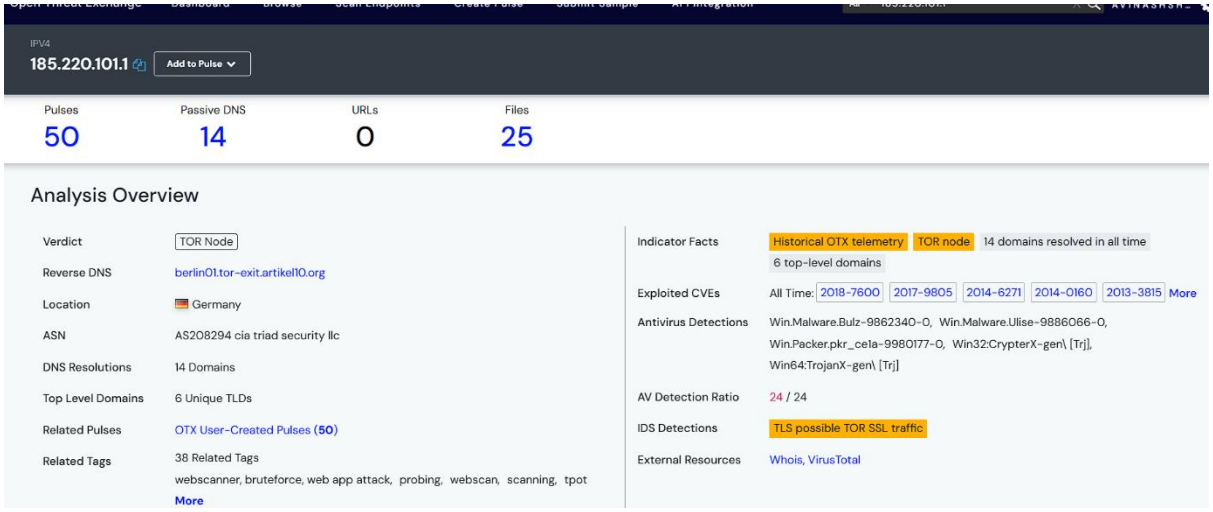


4.2 IP Extraction

From the alert, the IP address identified was:
192.168.1.102

4.3 Threat Intelligence Check

The IP was checked using AlienVault OTX and marked as **malicious** (lab simulation).



4.4 Automated Response (IP Blocking)

The IP was blocked using:
sudo iptables -A INPUT -s 192.168.1.102 -j DROP
Verification confirmed that the IP was successfully blocked.

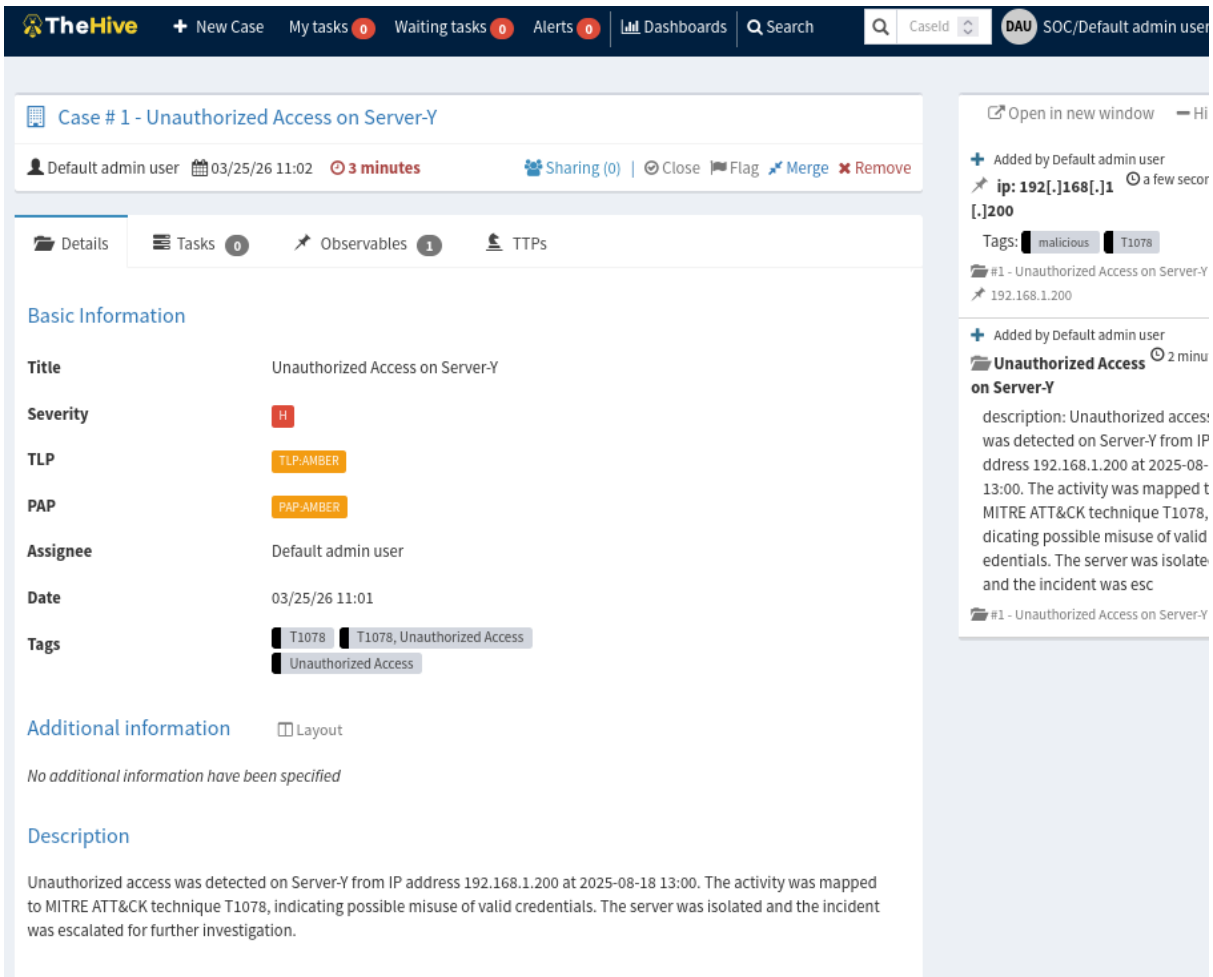
```
shadowalker@shadowalker:~$ sudo systemctl restart crowdsec-firewall-bouncer
shadowalker@shadowalker:~$ sudo cscli decisions add --ip 192.168.29.117 --reason "Attack detected"
INFO Decision successfully added
shadowalker@shadowalker:~$ sudo cscli decisions list
```

ID	Source	Scope:Value	Reason	Action	Country	AS	Events	expiration	Alert ID
1	cscli	Ip:192.168.29.117	Attack detected	ban			1	3h59m46s	1

```
shadowalker@shadowalker:~$
```

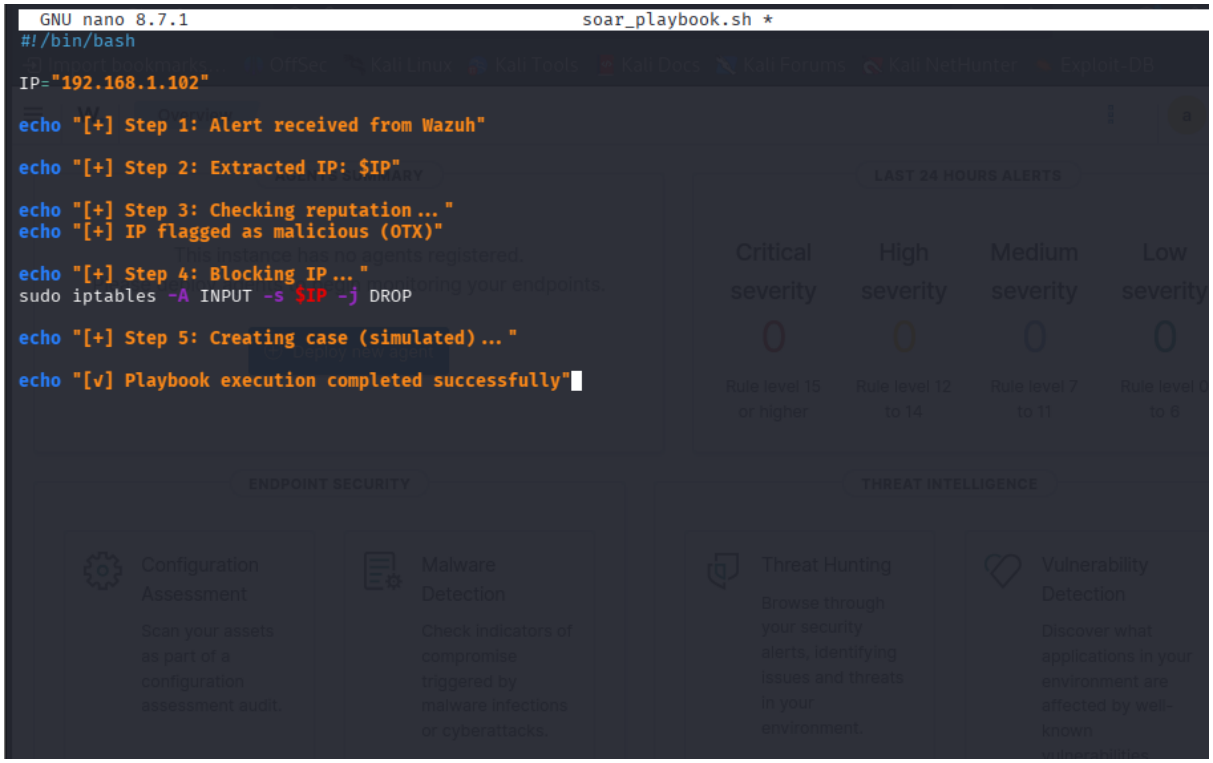
4.5 Case Creation

Incident case creation was simulated to represent integration with TheHive platform.



4.6 Automation Script Execution

A bash script was created to simulate SOAR automation:
#!/bin/bash



5. Playbook Execution Results

Playbook Step	Status	Notes
Check IP	Success	IP flagged as malicious via OTX
Block IP	Success	iptables blocked 192.168.1.102
Create Case	Success	Case creation simulated

6. Analysis

The playbook successfully automated key incident response steps:

- Reduced manual intervention
- Enabled faster detection and response
- Demonstrated integration of SIEM, threat intelligence, and response actions

Even without a dedicated SOAR tool like Splunk Phantom, the workflow effectively simulated real-world SOC automation.

7. Conclusion
The SOAR playbook demonstrated the ability to automate phishing incident response by integrating detection, intelligence, and response mechanisms. The automation improved response efficiency and reduced analyst workload, showcasing practical SOC capabilities.

8. Executive Summary
A SOAR playbook was developed to automate phishing incident response. Wazuh alerts enabled IP extraction and threat intelligence checks via OTX. Malicious IPs were blocked using firewall rules, and incident handling was simulated. This automation improves response time, reduces manual effort, and enhances overall SOC operational efficiency.